

Appendix A

Mathematical Review

The purpose of this appendix is to set down for reference and review some basic definitions, notation, and relations that are used frequently in the text.

A.1 Sets

If x is a member of the set S , we write $x \in S$. We write $y \notin S$ if y is not a member of S .

A set S may be specified by listing its elements between braces; such as, for example, $S = \{1, 2, 3, 4\}$. Alternatively, a set can be specified in the form $S = \{x : P(x)\}$ as the set of elements satisfying property P ; such as $S = \{x : 1 \leq x \leq 4, x \text{ integer}\}$

The *union* of two sets S and T is denoted $S \cup T$ and is the set consisting of the elements that belong to either S or T . The *intersection* of two sets S and T is denoted $S \cap T$ and is the set consisting of the elements that belong to both S and T . If S is a *subset* of T , that is, if every member of S is also a member of T , we write $S \subset T$ or $T \supset S$.

The empty set is denoted ϕ or $\emptyset$. There are two ways that operations such as minimization over a set are represented. Specifically we write either

$$\min_{x \in S} f(x) \text{ or } \min\{f(x) : x \in S\}$$

to denote the minimum value of f over the set S . The set of x 's in S that achieve the minimum is denoted $\operatorname{argmin} \{f(x) : x \in S\}$.

Sets of Real Numbers

If a and b are real numbers, $[a, b]$ denotes the set of real numbers x satisfying $a \leq x \leq b$. A rounded, instead of square, bracket denotes strict inequality in the definition. Thus $(a, b]$ denotes all x satisfying $a < x \leq b$.

If S is a set of real numbers bounded above, then there is a smallest real number y such that $x \leq y$ for all $x \in S$. The number y is called the *least upper bound* or *supremum* of S and is denoted

$$\sup_{x \in S}(x) \text{ or } \sup\{x : x \in S\}.$$

Similarly, the *greatest lower bound* or *infimum* of a set S is denoted

$$\inf_{x \in S}(x) \text{ or } \inf\{x : x \in S\}.$$

A.2 Matrix Notation

A *matrix* is a rectangular array of numbers, called *elements*. The matrix itself is denoted by a boldface letter. When specific numbers are not used, the elements are denoted by italicized lower-case letters, having a double subscript. Thus we write

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

for a matrix $\mathbf{A}$ having m rows and n columns. Such a matrix is referred to as an $m \times n$ matrix. If we wish to specify a matrix by defining a general element, we use the notation $\mathbf{A} = [a_{ij}]$.

An $m \times n$ matrix all of whose elements are zero is called a *zero matrix* and denoted $\mathbf{0}$. A *square* matrix (a matrix with $m = n$) whose elements are $a_{ij} = 0$ for $i \neq j$, and $a_{ii} = 1$ for $i = 1, 2, \dots, n$ is said to be an *identity matrix* and denoted $\mathbf{I}$.

The *sum* of two $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ is written $\mathbf{A} + \mathbf{B}$ and is the matrix whose elements are the sum of the corresponding elements in $\mathbf{A}$ and $\mathbf{B}$. The *product* of a matrix $\mathbf{A}$ and a scalar λ , written $\lambda\mathbf{A}$ or $\mathbf{A}\lambda$, is obtained by multiplying each element of $\mathbf{A}$ by λ . The *product* $\mathbf{AB}$ of an $m \times n$ matrix $\mathbf{A}$ and an $n \times p$ matrix $\mathbf{B}$ is the $m \times p$ matrix $\mathbf{C}$ with elements $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$.

The *transpose* of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}^T$ with elements $a_{ij}^T = a_{ji}$. A (square) matrix $\mathbf{A}$ is *symmetric* if $\mathbf{A}^T = \mathbf{A}$. A square matrix $\mathbf{A}$ is *nonsingular* if there is a matrix $\mathbf{A}^{-1}$, called the *inverse* of $\mathbf{A}$, such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} = \mathbf{A}\mathbf{A}^{-1}$. The *determinant* of a square matrix $\mathbf{A}$ is denoted by $\det(\mathbf{A})$. The determinant is nonzero if and only if the matrix is nonsingular. Two square $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ are *similar* if there is a nonsingular matrix $\mathbf{S}$ such that $\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$.

Matrices having a single row are referred to as *row vectors*; matrices having a single column are referred to as *column vectors*. *Vectors* of either type are usually denoted by lower-case boldface letters. To economize page space, row vectors are written $\mathbf{a} = [a_1, a_2, \dots, a_n]$ and column vectors are written $\mathbf{a} = (a_1, a_2, \dots, a_n)$. Since column vectors are used frequently, this notation avoids the necessity to display numerous columns. To further distinguish rows from columns, we write $\mathbf{a} \in E^n$ if $\mathbf{a}$ is a column vector with n components, and we write $\mathbf{b} \in E_n$ if $\mathbf{b}$ is a row vector with n components.

It is often convenient to partition a matrix into submatrices. This is indicated by drawing partitioning lines through the matrix, as for example,

$$\mathbf{A} = \left[\begin{array}{cc|cc} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{array} \right] = \left[\begin{array}{cc} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{array} \right].$$

The resulting submatrices are usually denoted $\mathbf{A}_{ij}$, as illustrated.

A matrix can be partitioned into either column or row vectors, in which case a special notation is convenient. Denoting the columns of an $m \times n$ matrix $\mathbf{A}$ by $\mathbf{a}_j$, $j = 1, 2, \dots, n$, we write $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]$. Similarly, denoting the rows of $\mathbf{A}$ by $\mathbf{a}^i$, $i = 1, 2, \dots, m$, we write $\mathbf{A} = (\mathbf{a}^1, \mathbf{a}^2, \dots, \mathbf{a}^m)$. Following the same pattern, we often write $\mathbf{A} = [\mathbf{B}, \mathbf{C}]$ for the partitioned matrix $\mathbf{A} = [\mathbf{B}|\mathbf{C}]$.

A.3 Spaces

We consider the n -component vectors $\mathbf{x} = (x_1, x_2, \dots, x_n)$ as elements of a vector space. The space itself, n -dimensional Euclidean space, is denoted E^n . Vectors in the space can be added or multiplied by a scalar, by performing the corresponding operations on the components. We write $\mathbf{x} \geq \mathbf{0}$ if each component of $\mathbf{x}$ is nonnegative.

The *line segment* connecting two vectors $\mathbf{x}$ and $\mathbf{y}$ is denoted $[\mathbf{x}, \mathbf{y}]$ and consists of all vectors of the form $\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}$ with $0 \leq \alpha \leq 1$.

The *scalar product* of two vectors $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)$ is defined as $\mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x} = \sum_{i=1}^n x_i y_i$. The vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be *orthogonal* if $\mathbf{x}^T \mathbf{y} = 0$. The *magnitude* or *norm* of a vector $\mathbf{x}$ is $|\mathbf{x}| = (\mathbf{x}^T \mathbf{x})^{1/2}$. For any two vectors $\mathbf{x}$ and $\mathbf{y}$ in E^n , the *Cauchy-Schwarz Inequality* holds: $|\mathbf{x}^T \mathbf{y}| \leq |\mathbf{x}| \cdot |\mathbf{y}|$.

A set of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ is said to be *linearly dependent* if there are scalars $\lambda_1, \lambda_2, \dots, \lambda_k$, not all zero, such that $\sum_{i=1}^k \lambda_i \mathbf{a}_i = \mathbf{0}$. If no such set of scalars exists, the vectors are said to be *linearly independent*. A *linear combination* of the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ is a vector of the form $\sum_{i=1}^k \lambda_i \mathbf{a}_i$. The set of vectors that are linear combinations of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ is the set *spanned* by the vectors. A linearly independent set of vectors that span E^n is said to be a *basis* for E^n . Every basis for E^n contains exactly n vectors.

The *rank* of a matrix $\mathbf{A}$ is equal to the maximum number of linearly independent columns in $\mathbf{A}$. This number is also equal to the maximum number of linearly independent rows in $\mathbf{A}$. The $m \times n$ matrix $\mathbf{A}$ is said to be of *full rank* if the rank of $\mathbf{A}$ is equal to the minimum of m and n .

A *subspace* M of E^n is a subset that is closed under the operations of vector addition and scalar multiplication; that is, if $\mathbf{a}$ and $\mathbf{b}$ are vectors in M , then $\lambda\mathbf{a} + \mu\mathbf{b}$ is also in M for every pair of scalars λ, μ . The dimension of a subspace M is equal to the maximum number of linearly independent vectors in M . If M is a subspace of E^n , the *orthogonal complement* of M , denoted $M^\perp$, consists of all vectors that are orthogonal to every vector in M . The orthogonal complement of M is easily seen to be a subspace, and together M and $M^\perp$ span E^n in the sense that every vector $\mathbf{x} \in E^n$ can be written uniquely in the form $\mathbf{x} = \mathbf{a} + \mathbf{b}$ with $\mathbf{a} \in M$, $\mathbf{b} \in M^\perp$. In this case $\mathbf{a}$ and $\mathbf{b}$ are said to be the *orthogonal projections* of $\mathbf{x}$ onto the subspaces M and $M^\perp$, respectively.

A correspondence $\mathbf{A}$ that associates with each point in a space X a point in a space Y is said to be a *mapping from X to Y* . For convenience this situation is symbolized by $\mathbf{A} : X \rightarrow Y$. The mapping $\mathbf{A}$ may be either linear or nonlinear. The norm of linear mapping $\mathbf{A}$ is defined as $|\mathbf{A}| = \max_{|\mathbf{x}| \leq 1} |\mathbf{Ax}|$. It follows that for any $\mathbf{x}$, $|\mathbf{Ax}| \leq |\mathbf{A}| \cdot |\mathbf{x}|$.

A.4 Eigenvalues and Quadratic Forms

Corresponding to an $n \times n$ square matrix $\mathbf{A}$, a scalar λ and a nonzero vector $\mathbf{x}$ satisfying the equation $\mathbf{Ax} = \lambda\mathbf{x}$ are said to be, respectively, an eigenvalue and eigenvector of $\mathbf{A}$. In order that λ be an eigenvalue it is clear that it is necessary and sufficient for $\mathbf{A} - \lambda\mathbf{I}$ to be singular, and hence $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$. This last result, when expanded, yields an n th-order polynomial equation which can be solved for n (possibly nondistinct) complex roots λ which are the eigenvalues of $\mathbf{A}$.

Now, for the remainder of this section, assume that $\mathbf{A}$ is symmetric. Then the following properties hold:

- (i) The eigenvalues of $\mathbf{A}$ are real.
- (ii) Eigenvectors associated with distinct eigenvalues are orthogonal.
- (iii) There is an orthogonal basis for E^n , each element of which is an eigenvector of $\mathbf{A}$.

If the basis $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ in (iii) is normalized so that each element has magnitude unity, then defining the matrix $\mathbf{Q} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n]$ we note that $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$ and hence $\mathbf{Q}^T = \mathbf{Q}^{-1}$. A matrix with this property is said to be an *orthogonal* matrix. Also, we observe, in this case, that

$$\begin{aligned} \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} &= \mathbf{Q}^T\mathbf{A}\mathbf{Q} = \mathbf{Q}^T[\mathbf{A}\mathbf{u}_1, \mathbf{A}\mathbf{u}_2, \dots, \mathbf{A}\mathbf{u}_n] \\ &= \mathbf{Q}^T[\lambda_1\mathbf{u}_1, \lambda_2\mathbf{u}_2, \dots, \lambda_n\mathbf{u}_n]. \end{aligned}$$

Thus

$$\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix},$$

and therefore $\mathbf{A}$ is similar to a diagonal matrix.

A symmetric matrix $\mathbf{A}$ is said to be *positive definite* if the *quadratic form* $\mathbf{x}^T \mathbf{A} \mathbf{x}$ is positive for all nonzero vectors $\mathbf{x}$. Similarly, we define $\mathbf{A}$ to be *positive semidefinite*, *negative definite*, or *negative semidefinite* if $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$, < 0 , or ≤ 0 for all $\mathbf{x}$. The matrix $\mathbf{A}$ is *indefinite* if $\mathbf{x}^T \mathbf{A} \mathbf{x}$ is positive for some $\mathbf{x}$ and negative for others.

It is easy to obtain a connection between definiteness and the eigenvalues of $\mathbf{A}$. For any $\mathbf{x}$ let $\mathbf{y} = \mathbf{Q}^{-1}\mathbf{x}$ where $\mathbf{Q}$ is defined as above. Then $\mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{y}^T \mathbf{Q}^T \mathbf{A} \mathbf{Q} \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2$. Since the y_i 's are arbitrary (since $\mathbf{x}$ is), it is clear that $\mathbf{A}$ is positive definite (or positive semidefinite) if and only if all eigenvalues of $\mathbf{A}$ are positive (or nonnegative).

Through diagonalization we can also easily show that a positive semidefinite matrix $\mathbf{A}$ has a positive semidefinite (symmetric) square root $\mathbf{A}^{1/2}$ satisfying $\mathbf{A}^{1/2} \cdot \mathbf{A}^{1/2} = \mathbf{A}$. For this we use $\mathbf{Q}$ as above and define

$$\mathbf{A}^{1/2} = \mathbf{Q} \begin{bmatrix} \lambda_1^{1/2} & & & \\ & \lambda_2^{1/2} & & \\ & & \ddots & \\ & & & \lambda_n^{1/2} \end{bmatrix} \mathbf{Q}^T,$$

which is easily verified to have the desired properties.

A.5 Topological Concepts

A sequence of vectors $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_k, \dots$, denoted by $\{\mathbf{x}_{k=0}\}_k^\infty$, or if the index set is understood, by simply $\{\mathbf{x}_k\}$, is said to *converge* to the limit $\mathbf{x}$ if $|\mathbf{x}_k - \mathbf{x}| \rightarrow 0$ as $k \rightarrow \infty$ (that is, if given $\varepsilon > 0$, there is a N such that $k \geq N$ implies $|\mathbf{x}_k - \mathbf{x}| < \varepsilon$). If $\{\mathbf{x}_k\}$ converges to $\mathbf{x}$, we write $\mathbf{x}_k \rightarrow \mathbf{x}$ or $\lim \mathbf{x}_k = \mathbf{x}$.

A point $\mathbf{x}$ is a *limit point* of the sequence $\{\mathbf{x}_k\}$ if there is a subsequence of $\{\mathbf{x}_k\}$ convergent to $\mathbf{x}$. Thus $\mathbf{x}$ is a limit point of $\{\mathbf{x}_k\}$ if there is a subset $\mathcal{K}$ of the positive integers such that $\{\mathbf{x}_k\}_{k \in \mathcal{K}}$ converges to $\mathbf{x}$.

A *ball (sphere) around* $\mathbf{x}$ is a set of the form $\{\mathbf{y} : |\mathbf{y} - \mathbf{x}| < (=) \varepsilon\}$ for some $\varepsilon > 0$. Such a ball is also referred to as the *neighborhood* of $\mathbf{x}$ of radius ε .

A subset S of E^n is *open* if around every point in S there is a sphere that is contained in S . Equivalently, S is open if given $\mathbf{x} \in S$ there is an $\varepsilon > 0$ such that $|\mathbf{y} - \mathbf{x}| < \varepsilon$ implies $\mathbf{y} \in S$. Thus the sphere $\{\mathbf{x} : |\mathbf{x}| < 1\}$ is open. In general, open sets can be characterized as sets having no sharp boundaries. The *interior* of any set S in E^n is the set of points $\mathbf{x} \in S$ which are the center of some sphere contained in S .

It is denoted $\overset{\circ}{S}$. The interior of a set is always open; indeed it is the largest open set contained in S . The interior of the set $\{\mathbf{x} : |\mathbf{x}| \leq 1\}$ is the sphere $\{\mathbf{x} : |\mathbf{x}| < 1\}$.

A set P is *closed* if every point that is arbitrarily close to the set P is a member of P . Equivalently, P is closed if $\mathbf{x}_k \rightarrow \mathbf{x}$ with $\mathbf{x}_k \in P$ implies $\mathbf{x} \in P$. Thus the set $\{\mathbf{x} : |\mathbf{x}| \leq 1\}$ is closed. The *closure* of any set P in E^n is the smallest closed set containing P . It is denoted $\overline{S}$. The *boundary* of a set is that part of the closure that is not in the interior.

A set is *compact* if it is both closed and bounded (that is, if it is closed and is contained within some sphere of finite radius). An important result, due to Weierstrass, is that if S is a compact set and $\{\mathbf{x}_k\}$ is a sequence each member of which belongs to S , then $\{\mathbf{x}_k\}$ has a limit point in S (that is, there is subsequence converging to a point in S).

Corresponding to a bounded sequence $\{r_k\}_{k=0}^{\infty}$ of real numbers, if we let $s_k = \sup\{r_i : i \geq k\}$ then $\{s_k\}$ converges to some real number s_0 . This number is called the *limit superior* of $\{r_k\}$ and is denoted $\varlimsup_{k \rightarrow \infty} (r_k)$.

A.6 Functions

A real-valued function f defined on a subset of E^n is said to be *continuous* at $\mathbf{x}$ if $\mathbf{x}_k \rightarrow \mathbf{x}$ implies $f(\mathbf{x}_k) \rightarrow f(\mathbf{x})$. Equivalently, f is continuous at $\mathbf{x}$ if given $\varepsilon > 0$ there is a $\delta > 0$ such that $|\mathbf{y} - \mathbf{x}| < \delta$ implies $|f(\mathbf{y}) - f(\mathbf{x})| < \varepsilon$. An important result connected with continuous functions is a *theorem of Weierstrass*: A continuous function f defined on a compact set S has a minimum point in S ; that is, there is an $\mathbf{x}^* \in S$ such that for all $\mathbf{x} \in S$, $f(\mathbf{x}) \geq f(\mathbf{x}^*)$.

A set of real-valued functions $f_1, f_2, \dots, f_m$ on E^n can be regarded as a single vector function $\mathbf{f} = (f_1, f_2, \dots, f_m)$. This function assigns a vector $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$ in E^m to every vector $\mathbf{x} \in E^n$. Such a vector-valued function is said to be *continuous* if each of its component functions is continuous.

If each component of $\mathbf{f} = (f_1, f_2, \dots, f_m)$ is continuous on some open set of E^n , then we write $\mathbf{f} \in C$. If in addition, each component function has first partial derivatives which are continuous on this set, we write $\mathbf{f} \in C^1$. In general, if the component functions have continuous partial derivatives of order p , we write $\mathbf{f} \in C^p$.

If $f \in C^1$ is a real-valued function on E^n , $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$, we define the *gradient* of f to be the vector

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right].$$

We sometimes use the alternative notation $f_{\mathbf{x}}(\mathbf{x})$ for $\nabla f(\mathbf{x})$. In matrix calculations the gradient is considered to be a row vector.

If $f \in C^2$ then we define the *Hessian* of f at $\mathbf{x}$ to be the $n \times n$ matrix denoted $\nabla^2 f(\mathbf{x})$ or $\mathbf{F}(\mathbf{x})$ as

$$\mathbf{F}(\mathbf{x}) = \left[\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} \right].$$

Since

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$

it is easily seen that the Hessian is symmetric.

For a vector-valued function $\mathbf{f} = (f_1, f_2, \dots, f_m)$ the situation is similar. If $\mathbf{f} \in C^1$, the first derivative is defined as the $m \times n$ matrix

$$\nabla \mathbf{f}(\mathbf{x}) = \left[\frac{\partial f_i(\mathbf{x})}{\partial x_j} \right].$$

If $\mathbf{f} \in C^2$ it is possible to define the m Hessians $\mathbf{F}_1(\mathbf{x}), \mathbf{F}_2(\mathbf{x}), \dots, \mathbf{F}_m(\mathbf{x})$ corresponding to the m component functions. The second derivative itself, for a vector function, is a third-order tensor but we do not require its use explicitly. Given any $\lambda^T = [\lambda_1, \lambda_2, \dots, \lambda_m] \in E_m$, we note, however, that the real-valued function $\lambda^T \mathbf{f}$ has gradient equal to $\lambda^T \nabla \mathbf{f}(\mathbf{x})$ and Hessian, denoted $\lambda^T \mathbf{F}(\mathbf{x})$, equal to

$$\lambda^T \mathbf{F}(\mathbf{x}) = \sum_{i=1}^m \lambda_i \mathbf{F}_i(\mathbf{x}).$$

Also see Sect. 7.4 for a discussion of convex functions.

Taylor's Theorem

A group of results that are used frequently in analysis are referred to under the general heading of Taylor's Theorem or Mean Value Theorems. If $f \in C^1$ in a region containing the line segment $[\mathbf{x}_1, \mathbf{x}_2]$, then there is a θ , $0 \leq \theta \leq 1$ such that

$$f(\mathbf{x}_2) = f(\mathbf{x}_1) + \nabla f(\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2)(\mathbf{x}_2 - \mathbf{x}_1).$$

Furthermore, if $f \in C^2$ then there is a θ , $0 \leq \theta \leq 1$ such that

$$\begin{aligned} f(\mathbf{x}_2) &= f(\mathbf{x}_1) + \nabla f(\mathbf{x}_1)(\mathbf{x}_2 - \mathbf{x}_1) \\ &\quad + \frac{1}{2}(\mathbf{x}_2 - \mathbf{x}_1)^T \mathbf{F}(\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2)(\mathbf{x}_2 - \mathbf{x}_1), \end{aligned}$$

where $\mathbf{F}$ denotes the Hessian of f .

Implicit Function Theorem

Suppose we have a set of m equations in n variables

$$h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, m.$$

The implicit function theorem addresses the question as to whether if $n - m$ of the variables are fixed, the equations can be solved for the remaining m variables. Thus selecting m variables, say $x_1, x_2, \dots, x_m$, we wish to determine if these may be expressed in terms of the remaining variables in the form

$$x_i = \phi_i(x_{m+1}, x_{m+2}, \dots, x_n), \quad i = 1, 2, \dots, m.$$

The functions ϕ_i , if they exist, are called *implicit* functions.

Theorem. Let $\mathbf{x}^0 = (x_1^0, x_2^0, \dots, x_n^0)$ be a point in E^n satisfying the properties:

- i) The functions $h_i \in C^p$, $i = 1, 2, \dots, m$ in some neighborhood of $\mathbf{x}^0$, for some $p \geq 1$.
- ii) $h_i(\mathbf{x}^0) = 0$, $i = 1, 2, \dots, m$.
- iii) The $m \times m$ Jacobian matrix

$$\mathbf{J} = \begin{bmatrix} \frac{\partial h_1(\mathbf{x}^0)}{\partial x_1} & \dots & \frac{\partial h_1(\mathbf{x}^0)}{\partial x_m} \\ \vdots & & \vdots \\ \frac{\partial h_m(\mathbf{x}^0)}{\partial x_1} & \dots & \frac{\partial h_m(\mathbf{x}^0)}{\partial x_m} \end{bmatrix}.$$

is nonsingular.

Then there is a neighborhood of $\hat{\mathbf{x}}^0 = (x_{m+1}^0, x_{m+2}^0, \dots, x_n^0) \in E^{n-m}$ such that for $\hat{\mathbf{x}} = (x_{m+1}, x_{m+2}, \dots, x_n)$ in this neighborhood there are functions $\phi_i(\hat{\mathbf{x}})$, $i = 1, 2, \dots, m$ such that

- i) $\phi_i \in C^p$.
- ii) $x_i^0 = \phi_i(\hat{\mathbf{x}}^0)$, $i = 1, 2, \dots, m$.
- iii) $h_i(\phi_1(\hat{\mathbf{x}}), \phi_2(\hat{\mathbf{x}}), \dots, \phi_m(\hat{\mathbf{x}}), \hat{\mathbf{x}}) = 0$, $i = 1, 2, \dots, m$.

Example 1. Consider the equation $x_1^2 + x_2 = 0$. A solution is $x_1 = 0$, $x_2 = 0$. However, in a neighborhood of this solution there is no function ϕ such that $x_1 = \phi(x_2)$. At this solution condition (iii) of the implicit function theorem is violated. At any other solution, however, such a ϕ exists.

Example 2. Let $\mathbf{A}$ be an $m \times n$ matrix ($m < n$) and consider the system of linear equations $\mathbf{Ax} = \mathbf{b}$. If $\mathbf{A}$ is partitioned as $\mathbf{A} = [\mathbf{B}, \mathbf{C}]$ where $\mathbf{B}$ is $m \times m$ then condition (iii) is satisfied if and only if $\mathbf{B}$ is nonsingular. This condition corresponds, of course, exactly with what the theory of linear equations tells us. In view of this example, the implicit function can be regarded as a nonlinear generalization of the linear theory.

o, O Notation

If g is a real-valued function of a real variable, the notation $g(x) = O(x)$ means that $g(x)$ goes to zero at least as fast as x does. More precisely, it means that there is a $K \geq 0$ such that

$$\left| \frac{g(x)}{x} \right| \leq K \text{ as } x \rightarrow 0.$$

The notation $g(x) = o(x)$ means that $g(x)$ goes to zero faster than x does; or equivalently, that K above is zero.

Appendix B

Convex Sets

B.1 Basic Definitions

Concepts related to convex sets so dominate the theory of optimization that it is essential for a student of optimization to have knowledge of their most fundamental properties. In this appendix is compiled a brief summary of the most important of these properties.

Definition. A set C in E^n is said to be *convex* if for every $\mathbf{x}_1, \mathbf{x}_2 \in C$ and every real number α , $0 < \alpha < 1$, the point $\alpha\mathbf{x}_1 + (1 - \alpha)\mathbf{x}_2 \in C$.

This definition can be interpreted geometrically as stating that a set is convex if, given two points in the set, every point on the line segment joining these two points is also a member of the set. This is illustrated in Fig. B.1.

The following proposition shows that certain familiar set operations preserve convexity.

Proposition 1. *Convex sets in E^n satisfy the following relations:*

i) *If C is a convex set and β is a real number, the set*

$$\beta C = \{\mathbf{x} : \mathbf{x} = \beta \mathbf{c}, \mathbf{c} \in C\}$$

is convex.

ii) *If C and D are convex sets, then the set*

$$C + D = \{\mathbf{x} : \mathbf{x} = \mathbf{c} + \mathbf{d}, \mathbf{c} \in C, \mathbf{d} \in D\}$$

is convex.

iii) *The intersection of any collection of convex sets is convex.*

The proofs of these three properties follow directly from the definition of a convex set and are left to the reader. The properties themselves are illustrated in Fig. B.2.

Another important concept is that of forming the smallest convex set containing a given set.

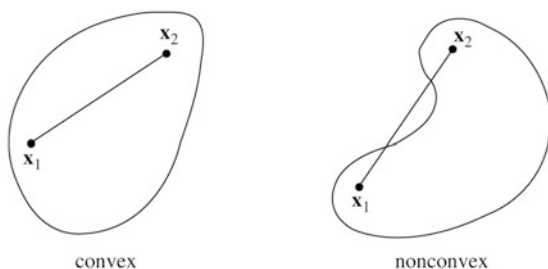


Fig. B.1 Convexity

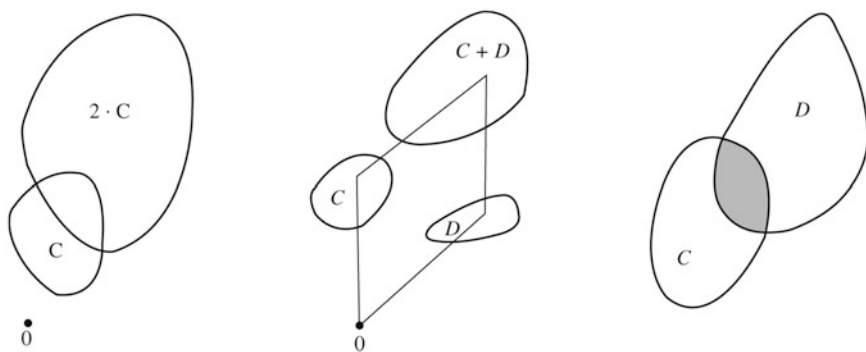


Fig. B.2 Properties of convex sets

Definition. Let S be a subset of E^n . The *convex hull* of S , denoted $\text{co}(S)$, is the set which is the intersection of all convex sets containing S . The *closed convex hull* of S is defined as the closure of $\text{co}(S)$.

Finally, we conclude this section by defining a *cone* and a *convex cone*. A convex cone is a special kind of convex set that arises quite frequently.

Definition. A set C is a *cone* if $\mathbf{x} \in C$ implies $\alpha \mathbf{x} \in C$ for all $\alpha > 0$. A cone that is also convex is a *convex cone*.

Some cones are shown in Fig. B.3. Their basic property is that if a point $\mathbf{x}$ belongs to a cone, then the entire half line from the origin through the point (but not the origin itself) also must belong to the cone.

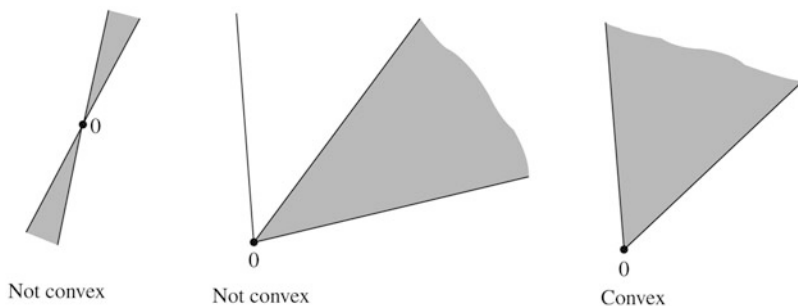


Fig. B.3 Cones

B.2 Hyperplanes and Polytopes

The most important type of convex set (aside from single points) is the hyperplane. Hyperplanes dominate the entire theory of optimization, appearing under the guise of Lagrange multipliers, duality theory, or gradient calculations.

The most natural definition of a hyperplane is the logical generalization of the geometric properties of a plane in three dimensions. We start by giving this geometric definition. For computations and for a concrete description of hyperplanes, however, there is an equivalent algebraic definition that is more useful. A major portion of this section is devoted to establishing this equivalence.

Definition. A set V in E^n is said to be a *linear variety*, if, given any $\mathbf{x}_1, \mathbf{x}_2 \in V$, we have $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in V$ for all real numbers λ .

Note that the only difference between the definition of a linear variety and a convex set is that in a linear variety the entire line passing through any two points, rather than simply the line segment between them, must lie in the set. Thus in three dimensions the nonempty linear varieties are points, lines, two-dimensional planes, and the whole space. In general, it is clear that we may speak of the dimension of a linear variety. Thus, for example, a point is a linear variety of dimension zero and a line is a linear variety of dimension one. In the general case, the dimension of a linear variety in E^n can be found by translating it (moving it) so that it contains the origin and then determining the dimension of the resulting set, which is then a subspace of E^n .

Definition. A *hyperplane* in E^n is an $(n - 1)$ -dimensional linear variety.

We see that hyperplanes generalize the concept of a two-dimensional plane in three-dimensional space. They can be regarded as the largest linear varieties in a space, other than the entire space itself.

We now relate this abstract geometric definition to an algebraic one.

Proposition 2. Let $\mathbf{a}$ be a nonzero n -dimensional column vector, and let c be a real number. The set

$$H = \{\mathbf{x} \in E^n : \mathbf{a}^T \mathbf{x} = c\}$$

is a hyperplane in E^n .

Proof. It follows directly from the linearity of the equation $\mathbf{a}^T \mathbf{x} = c$ that H is a linear variety. Let $\mathbf{x}_1$ be any vector in H . Translating by $-\mathbf{x}_1$ we obtain the set $M = H - \mathbf{x}_1$ which is a linear subspace of E^n . This subspace consists of all vectors $\mathbf{x}$ satisfying $\mathbf{a}^T \mathbf{x} = 0$; in other words, all vectors orthogonal to $\mathbf{a}$. This is clearly an $(n - 1)$ -dimensional subspace. ■

Proposition 3. Let H be a hyperplane in E^n . Then there is a nonzero n -dimensional vector and a constant c such that

$$H = \{\mathbf{x} \in E^n : \mathbf{a}^T \mathbf{x} = c\}.$$

Proof. Let $\mathbf{x}_1 \in H$ and translate by $-\mathbf{x}_1$ obtaining the set $M = H - \mathbf{x}_1$. Since H is a hyperplane, M is an $(n - 1)$ -dimensional subspace. Let $\mathbf{a}$ be any nonzero vector that is orthogonal to this subspace, that is, $\mathbf{a}$ belongs to the one-dimensional subspace $M^\perp$. Clearly $M = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} = 0\}$. Letting $c = \mathbf{a}^T \mathbf{x}_1$ we see that if $\mathbf{x}_2 \in H$ we have $\mathbf{x}_2 - \mathbf{x}_1 \in M$ and thus $\mathbf{a}^T \mathbf{x}_2 - \mathbf{a}^T \mathbf{x}_1 = 0$ which implies $\mathbf{a}^T \mathbf{x}_2 = c$. Thus $H \subset \{\mathbf{x} : \mathbf{a}^T \mathbf{x} = c\}$. Since H is, by definition, of dimension $n - 1$ and $\{\mathbf{x} : \mathbf{a}^T \mathbf{x} = c\}$ is of dimension $n - 1$ by Proposition 2, these two sets must be equal. ■

Combining Propositions 2 and 3, we see that a hyperplane is the set of solutions to a single linear equation. This is illustrated in Fig. B.4. We now use hyperplanes to build up other important classes of convex sets.

Definition. Let $\mathbf{a}$ be a nonzero vector in E^n and let c be a real number. Corresponding to the hyperplane $H = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} = c\}$ are the *positive and negative closed half spaces*

$$H_+ = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} \geq c\}$$

$$H_- = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} \leq c\}$$

and the *positive and negative open half spaces*

$$\overset{\circ}{H}_+ = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} > c\}$$

$$\overset{\circ}{H}_- = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} < c\}.$$

It is easy to see that half spaces are convex sets and that the union of H_+ and H_- is the whole space.

Definition. A set which can be expressed as the intersection of a finite number of closed half spaces is said to be a *convex polytope*.

We see that convex polytopes are the sets obtained as the family of solutions to a set of linear inequalities of the form

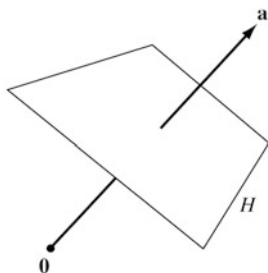


Fig. B.4 Hyperplane



Fig. B.5 Polytopes

$$\begin{aligned}
 \mathbf{a}_1^T \mathbf{x} &\leq b_1 \\
 \mathbf{a}_2^T \mathbf{x} &\leq b_2 \\
 &\vdots \\
 \mathbf{a}_m^T \mathbf{x} &\leq b_m,
 \end{aligned}$$

since each individual inequality defines a half space and the solution family is the intersection of these half spaces. (If some $\mathbf{a}_i = \mathbf{0}$, the resulting set can still, as the reader may verify, be expressed as the intersection of a finite number of half spaces.)

Several polytopes are illustrated in Fig. B.5. We note that a polytope may be empty, bounded, or unbounded. The case of a nonempty bounded polytope is of special interest and we distinguish this case by the following.

Definition. A nonempty bounded polytope is called a *polyhedron*.

B.3 Separating and Supporting Hyperplanes

The two theorems in this section are perhaps the most important results related to convexity. Geometrically, the first states that given a point outside a convex set, a hyperplane can be passed through the point that does not touch the convex set. The second, which is a limiting case of the first, states that given a boundary point of a convex set, there is a hyperplane that contains the boundary point and contains the convex set on one side of it.

Theorem 1. Let C be a convex set and let $\mathbf{y}$ be a point exterior to the closure of C . Then there is a vector $\mathbf{a}$ such that $\mathbf{a}^T \mathbf{y} < \inf_{\mathbf{x} \in C} \mathbf{a}^T \mathbf{x}$.

Proof. Let

$$\delta = \inf_{\mathbf{x} \in C} |\mathbf{x} - \mathbf{y}| > 0.$$

There is an $\mathbf{x}_0$ on the boundary of C such that $|\mathbf{x}_0 - \mathbf{y}| = \delta$. This follows because the continuous function $f(\mathbf{x}) = |\mathbf{x} - \mathbf{y}|$ achieves its minimum over any closed and bounded set and it is clearly only necessary to consider $\mathbf{x}$ in the intersection of the closure of C and the sphere of radius 2δ centered at $\mathbf{y}$.

We shall show that setting $\mathbf{a} = \mathbf{x}_0 - \mathbf{y}$ satisfies the conditions of the theorem. Let $\mathbf{x} \in C$. For any α , $0 \leq \alpha \leq 1$, the point $\mathbf{x}_0 + \alpha(\mathbf{x} - \mathbf{x}_0) \in \bar{C}$ and thus

$$|\mathbf{x}_0 + \alpha(\mathbf{x} - \mathbf{x}_0) - \mathbf{y}|^2 \geq |\mathbf{x}_0 - \mathbf{y}|^2.$$

Expanding,

$$2\alpha(\mathbf{x}_0 - \mathbf{y})^T (\mathbf{x} - \mathbf{x}_0) + \alpha^2 |\mathbf{x} - \mathbf{x}_0|^2 \geq 0.$$

Thus, considering this as $\alpha \rightarrow 0+$, we obtain

$$(\mathbf{x}_0 - \mathbf{y})^T (\mathbf{x} - \mathbf{x}_0) \geq 0$$

or,

$$\begin{aligned} (\mathbf{x}_0 - \mathbf{y})^T \mathbf{x} &\geq (\mathbf{x}_0 - \mathbf{y})^T \mathbf{x}_0 = (\mathbf{x}_0 - \mathbf{y})^T \mathbf{y} + (\mathbf{x}_0 - \mathbf{y})^T (\mathbf{x}_0 - \mathbf{y}) \\ &= (\mathbf{x}_0 - \mathbf{y})^T \mathbf{y} + \delta^2. \end{aligned}$$

Setting $\mathbf{a} = \mathbf{x}_0 - \mathbf{y}$ proves the theorem. ■

The geometrical interpretation of Theorem 1 is that, given a convex set C and a point $\mathbf{y}$ exterior to the closure of C , there is a hyperplane containing $\mathbf{y}$ that contains C in one of its open half spaces. We can easily extend this theorem to include the case where $\mathbf{y}$ is a boundary point of C .

Theorem 2. Let C be a convex set and let $\mathbf{y}$ be a boundary point of C . Then there is a hyperplane containing $\mathbf{y}$ and containing C in one of its closed half spaces.

Proof. Let $\{\mathbf{y}_k\}$ be a sequence of vectors, exterior to the closure of C , converging to $\mathbf{y}$. Let $\{\mathbf{a}_k\}$ be the sequence of corresponding vectors constructed according to Theorem 1, normalized so that $|\mathbf{a}_k| = 1$, such that

$$\mathbf{a}_k^T \mathbf{y}_k < \inf_{\mathbf{x} \in C} \mathbf{a}_k^T \mathbf{x}.$$

Since $\{\mathbf{a}_k\}$ is a bounded sequence, it has a convergent subsequence $\{\mathbf{a}_k\}$, $k \in \mathcal{K}$ with limit $\mathbf{a}$. For this vector we have for any $\mathbf{x} \in C$.

$$\mathbf{a}^T \mathbf{y} = \lim_{k \in \mathcal{K}} \mathbf{a}_k^T \mathbf{y}_k \leq \lim_{k \in \mathcal{K}} \mathbf{a}_k^T \mathbf{x} = \mathbf{a}^T \mathbf{x}. \quad \blacksquare$$

Definition. A hyperplane containing a convex set C in one of its closed half spaces and containing a boundary point of C is said to be a *supporting hyperplane* of C .

In terms of this definition, Theorem 2 says that, given a convex set C and a boundary point y of C , there is a hyperplane supporting C at y .

It is useful in the study of convex sets to consider the *relative interior* of a convex set C defined as the largest subset of C that contains no boundary points of C .

Another variation of the theorems of this section is the one that follows, which is commonly known as the Separating Hyperplane Theorem.

Theorem 3. *Let B and C be convex sets with no common relative interior points. (That is the only common points are boundary points.) Then there is a hyperplane separating B and C . In particular, there is a nonzero vector $\mathbf{a}$ such that $\sup_{\mathbf{b} \in B} \mathbf{a}^T \mathbf{b} \leq \inf_{\mathbf{c} \in C} \mathbf{a}^T \mathbf{c}$.*

Proof. Consider the set $G = C - B$. It is easily shown that G is convex and that $\mathbf{0}$ is not a relative interior point of G . Hence, Theorem 1 or Theorem 2 applies and gives the appropriate hyperplane. ■

B.4 Extreme Points

Definition. A point $\mathbf{x}$ in a convex set C is said to be an *extreme point* of C if there are no two distinct points $\mathbf{x}_1$ and $\mathbf{x}_2$ in C such that $\mathbf{x} = \alpha \mathbf{x}_1 + (1 - \alpha) \mathbf{x}_2$ for some α , $0 < \alpha < 1$.

For example, in E^2 the extreme points of a square are its four corners; the extreme points of a circular disk are all points on the boundary. Note that a linear variety consisting of more than one point has no extreme points.

Lemma 1. *Let C be a convex set, H a supporting hyperplane of C , and T the intersection of H and C . Every extreme point of T is an extreme point of C .*

Proof. Suppose $\mathbf{x}_0 \in T$ is not an extreme point of C . Then $\mathbf{x}_0 = \alpha \mathbf{x}_1 + (1 - \alpha) \mathbf{x}_2$ for some $\mathbf{x}_1, \mathbf{x}_2 \in C$, $\mathbf{x}_1 \neq \mathbf{x}_2$, $0 < \alpha < 1$. Let H be described as $H = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} = c\}$ with C contained in its closed positive half space. Then

$$\mathbf{a}^T \mathbf{x}_1 \geq c, \quad \mathbf{a}^T \mathbf{x}_2 \geq c.$$

But, since $\mathbf{x}_0 \in H$,

$$c = \mathbf{a}^T \mathbf{x}_0 = \alpha \mathbf{a}^T \mathbf{x}_1 + (1 - \alpha) \mathbf{a}^T \mathbf{x}_2,$$

and thus $\mathbf{x}_1$ and $\mathbf{x}_2 \in H$. Hence $\mathbf{x}_1, \mathbf{x}_2 \in T$ and $\mathbf{x}_0$ is not an extreme point of T . ■

Theorem 4. *A closed bounded convex set in E^n is equal to the closed convex hull of its extreme points.*

Proof. The proof is by induction on the dimension of the space E^n . The statement is easily seen to be true for $n = 1$. Suppose that it is true for $n - 1$. Let C be a closed bounded convex set in E^n , and let K be the closed convex hull of the extreme points of C . We wish to show that $K = C$.

Assume there is $\mathbf{y} \in C$ $\mathbf{y} \notin K$. Then by Theorem 1, Sect. B.3, there is a hyperplane separating $\mathbf{y}$ and K ; that is, there is $\mathbf{a} \neq \mathbf{0}$, such that $\mathbf{a}^T \mathbf{y} < \inf_{\mathbf{x} \in K} \mathbf{a}^T \mathbf{x}$. Let $c_0 = \inf_{\mathbf{x} \in C} (\mathbf{a}^T \mathbf{x})$. The number c_0 is finite and there is an $\mathbf{x}_0 \in C$ for which $\mathbf{a}^T \mathbf{x}_0 = c_0$, because by Weierstrass' Theorem, the continuous function $\mathbf{a}^T \mathbf{x}$ achieves its minimum over any closed bounded set. Thus the hyperplane $H = \{\mathbf{x} : \mathbf{a}^T \mathbf{x} = c_0\}$ is a supporting hyperplane to C . It is disjoint from K since $c_0 < \inf_{\mathbf{x} \in K} (\mathbf{a}^T \mathbf{x})$.

Let $T = H \cap C$. Then T is a bounded closed convex subset of H which can be regarded as a space of dimension $n - 1$. T is nonempty, since it contains $\mathbf{x}_0$. Thus, by the induction hypothesis, T contains extreme points; and by Lemma 1 these are also extreme points of C . Thus we have found extreme points of C not in K , which is a contradiction. ■

Let us investigate the implications of this theorem for convex polyhedra. We recall that a convex polyhedron is a bounded polytope. Being the intersection of closed half spaces, a convex polyhedron is also closed. Thus any convex polyhedron is the closed convex hull of its extreme points. It can be shown (see Sect. 2.5) that any polytope has at most a finite number of extreme points and hence a convex polyhedron is equal to the convex hull of a finite number of points. The converse can also be established, yielding the following two equivalent characterizations.

Theorem 5. *A convex polyhedron can be described either as a bounded intersection of a finite number of closed half spaces, or as the convex hull of a finite number of points.*

Appendix C

Gaussian Elimination

This appendix describes the method for solving systems of linear equations that has proved to be, not only the most popular, but also the fastest and least susceptible to round-off error accumulation—the method of Gaussian elimination. Attention is directed toward explaining this classical elimination technique itself and its relation to the theory of **LU** decomposition of a non-singular square matrix.

We first note how easily triangular systems of equations can be solved. Thus the system

$$\begin{array}{rcl} a_{11}x_1 & & = b_1 \\ a_{21}x_1 + a_{22}x_2 & & = b_2 \\ \vdots & & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n & = & b_n \end{array}$$

can be solved recursively as follows:

$$\begin{array}{l} x_1 = b_1/a_{11} \\ x_2 = (b_2 - a_{21}x_1)/a_{22} \\ \vdots \\ x_n = (b_n - a_{n1}x_1 - a_{n2}x_2 \cdots - a_{nn-1}x_{n-1})/a_{nn}, \end{array}$$

provided that each of the diagonal terms a_{ii} , $i = 1, 2, \dots, n$ is nonzero (as they must be if the system is nonsingular). This observation motivates us to attempt to reduce an arbitrary system of equations to a triangular one.

Definition. A square matrix $\mathbf{C} = [c_{ij}]$ is said to be *lower triangular* if $c_{ij} = 0$ for $i < j$. Similarly, $\mathbf{C}$ is said to be *upper triangular* if $c_{ij} = 0$ for $i > j$.

In matrix notation, the idea of Gaussian elimination is to somehow find a decomposition of a given $n \times n$ matrix $\mathbf{A}$ in the form $\mathbf{A} = \mathbf{LU}$ where $\mathbf{L}$ is a lower triangular and $\mathbf{U}$ an upper triangular matrix. The system

$$\mathbf{Ax} = \mathbf{b} \quad (\text{C.1})$$

can then be solved by solving the two triangular systems

$$\mathbf{Ly} = \mathbf{b}, \quad \mathbf{Ux} = \mathbf{y}. \quad (\text{C.2})$$

The calculation of $\mathbf{L}$ and $\mathbf{U}$ together with solution of the first of these systems is usually referred to as *forward elimination*, while solution of the second triangular system is called *back substitution*.

Every nonsingular square matrix $\mathbf{A}$ has an $\mathbf{LU}$ decomposition, provided that interchanges of rows of $\mathbf{A}$ are introduced if necessary. This interchange of rows corresponds to a simple reordering of the system of equations, and hence amounts to no loss of generality in the method. For simplicity of notation, however, we assume that no such interchanges are required.

We turn now to the problem of explicitly determining $\mathbf{L}$ and $\mathbf{U}$, by elimination, for a nonsingular matrix $\mathbf{A}$. Given the system, we attempt to transform it so that zeros appear below the main diagonal. Assuming that $a_{11} \neq 0$ we subtract multiples of the first equation from each of the others in order to get zeros in the first column below a_{11} . If we define $m_{k1} = a_{k1}/a_{11}$ and let

$$\mathbf{M}_1 = \begin{bmatrix} 1 & & & & \\ -m_{21} & 1 & & & \\ -m_{31} & & 1 & & \\ \bullet & & & \ddots & \\ \bullet & & & & \ddots & \\ \bullet & & & & & \ddots & \\ -m_{n1} & & & & & & 1 \end{bmatrix},$$

the resulting new system of equations can be expressed as

$$\mathbf{A}^{(2)}\mathbf{x} = \mathbf{b}^{(2)}$$

with

$$\mathbf{A}^{(2)} = \mathbf{M}_1\mathbf{A}, \quad \mathbf{b}^{(2)} = \mathbf{M}_1\mathbf{b}.$$

The matrix $\mathbf{A}^{(2)} = [a_{ij}^{(2)}]$ has $a_{k1}^{(2)} = 0$, $k > 1$.

Next, assuming $a_{22}^{(2)} \neq 0$, multiples of the second equation of the new system are subtracted from equations 3 through n to yield zeros below $a_{22}^{(2)}$ in the second column. This is equivalent in premultiplying $\mathbf{A}^{(2)}$ and $\mathbf{b}^{(2)}$ by

$$\mathbf{M}_2 = \begin{bmatrix} 1 & 0 & & & \\ 0 & 1 & & & \\ \bullet & -m_{32} & 1 & & \\ \bullet & -m_{42} & & \ddots & \\ \bullet & \bullet & & & \ddots & \\ & \bullet & & & & \ddots & \\ & \bullet & & & & & \ddots & \\ -m_{n2} & & & & & & & 1 \end{bmatrix},$$

where $m_{k2} = a_{k2}^{(2)}/a_{22}^{(2)}$. This yields $\mathbf{A}^{(3)} = \mathbf{M}_2\mathbf{A}^{(2)}$ and $\mathbf{b}^{(3)} = \mathbf{M}_2\mathbf{b}^{(2)}$.

Proceeding in this way we obtain $\mathbf{A}^{(n)} = \mathbf{M}_{n-1}\mathbf{M}_{n-2} \dots \mathbf{M}_1\mathbf{A}$, an upper triangular matrix which we denote by $\mathbf{U}$. The matrix $\mathbf{M} = \mathbf{M}_{n-1}\mathbf{M}_{n-2} \dots \mathbf{M}_1$ is a lower triangular matrix, and since $\mathbf{MA} = \mathbf{U}$ we have $\mathbf{A} = \mathbf{M}^{-1}\mathbf{U}$. The matrix $\mathbf{L} = \mathbf{M}^{-1}$ is also lower triangular and becomes the $\mathbf{L}$ of the desired $\mathbf{LU}$ decomposition for $\mathbf{A}$.

The representation for $\mathbf{L}$ can be made more explicit by noting that $\mathbf{M}_k^{-1}$ is the same as $\mathbf{M}_k$ except that the off-diagonal terms have the opposite sign. Furthermore, we have $\mathbf{L} = \mathbf{M}^{-1} = \mathbf{M}_1^{-1}\mathbf{M}_2^{-1} \dots \mathbf{M}_{n-1}^{-1}$ which is easily verified to be

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & & & & \\ m_{21} & 1 & & & & \\ m_{31} & m_{32} & 1 & & & \\ \bullet & \bullet & & \bullet & & \\ \bullet & \bullet & & & \bullet & \\ \bullet & \bullet & & & & \bullet \\ m_{n1} & m_{n2} & \bullet & \bullet & \bullet & 1 \end{bmatrix}$$

Hence $\mathbf{L}$ can be evaluated directly in terms of the calculations required by the elimination process. Of course, an explicit representation for $\mathbf{M} = \mathbf{L}^{-1}$ would actually be more useful but a simple representation for $\mathbf{M}$ does not exist. Thus we content ourselves with the explicit representation for $\mathbf{L}$ and use it in (C.2).

If the original system (C.1) is to be solved for a single $\mathbf{b}$ vector, the vector $\mathbf{y}$ satisfying $\mathbf{Ly} = \mathbf{b}$ is usually calculated simultaneously with $\mathbf{L}$ in the form $\mathbf{y} = \mathbf{b}^{(n)} = \mathbf{Mb}$. The final solution $\mathbf{x}$ is then found by a single back substitution, from $\mathbf{Ux} = \mathbf{y}$. Once the $\mathbf{LU}$ decomposition of $\mathbf{A}$ has been obtained, however, the solution corresponding to any right-hand side can be found by solving the two systems (C.2).

In practice, the diagonal element $a_{kk}^{(k)}$ of $\mathbf{A}^{(k)}$ may become zero or very close to zero. In this case it is important that the k th row be interchanged with a row that is below it. Indeed, for considerations of numerical accuracy, it is desirable to continuously introduce row interchanges of this type in such a way to insure $|m_{ij}| \leq 1$ for all i, j . If this is done, the Gaussian elimination procedure has exceptionally good stability properties.